أنواع السحاب في زمن الضباب

Lecture 2 — Types of Clouds, Providers, and Data Centers

Q/ What are the two questions about any cloud ?

1. **Service model** — WHAT do you get? IaaS · PaaS · SaaS · DaaS (Lecture 1).
2. **Deployment model** — WHO owns the infrastructure and WHO shares it? Private · Public · Hybrid · Multi-cloud (Lecture 2).

The two choices are **independent**. Example: Gmail is **SaaS** (service model) delivered from a **public** cloud (deployment model).

Deployment models

> **Private Cloud** : cloud infrastructure **dedicated to a single organization**, run in its own data center or in a provider's dedicated space.

Q/ What are the characteristics of a private cloud ?

1. One tenant: no sharing with other customers.
2. Full control of hardware, network and policy.
3. High upfront cost (**CapEx**) and in-house staff.
4. Elasticity limited by the hardware you own.

Q/ Name technologies used to build a private cloud.

- OpenStack · VMware vSphere · Proxmox · Kubernetes on your own servers.

Security angle: you own the whole stack, so you own the **whole security burden** (physical access, patching, monitoring, incident response).

> **Public Cloud** : infrastructure **owned and operated by a provider** and rented to many customers (tenants) over the internet.

Q/ What are the characteristics of a public cloud ?

1. **Multi-tenant**: resources shared and isolated logically.
2. **Pay-as-you-go** (OpEx), no upfront hardware.
3. Near-instant elasticity and global regions.

4. The provider secures the infrastructure layers.

Q/ Name the major public cloud providers.

- AWS · Microsoft Azure · Google Cloud (also Alibaba, Oracle, IBM, DigitalOcean).

Security angle: **shared responsibility** — the provider secures the cloud, you secure what you put **in** it. **Misconfiguration** is a leading cause of public-cloud breaches.

Aspect	Private cloud	Public cloud
Ownership	Single organization	Provider; shared by many tenants
Cost model	High upfront (CapEx)	Pay-as-you-go (OpEx)
Elasticity	Limited by owned hardware	Near-unlimited, on demand
Control	Full	Limited to what the provider exposes
Time to deploy	Weeks to months	Minutes
Security duty	Entirely the organization	Shared with the provider
Best for	Strict compliance, steady workloads	Variable demand, fast growth

> **Hybrid Cloud** : a **private** cloud (or on-premises) and a **public** cloud working together, connected by a VPN or dedicated link.

Q/ What are the uses of hybrid cloud ?

1. **Cloud bursting**: run normally in-house; rent public capacity only for peaks.
2. **Data residency**: keep regulated data local; use the public cloud for everything else.
3. **Gradual migration**: move workloads to the cloud step by step.

Security angle: the **link between the two worlds** is a new attack surface; encrypt it and keep policies consistent on both sides.

> **Multi-Cloud** : using **two or more public cloud providers**, for different workloads or for the same workload.

Multi-cloud benefits	Multi-cloud challenges
Less lock-in	More skills needed

Multi-cloud benefits	Multi-cloud challenges
Resilience to one provider's outage	Different IAM and tools per provider
Best service for each task	Egress fees between clouds

Hybrid	Multi-cloud
Private + public, working together	Two or more **public** providers

A company can be **both** hybrid and multi-cloud at the same time.

Providers

> **Provider Lock-in** : switching providers becomes so costly or difficult that you stay even when you would rather leave.

Q/ What creates lock-in ?

1. Proprietary services and APIs (managed databases, serverless).
2. **Data gravity**: large datasets are slow and costly to move out.
3. **Egress** (data-out) fees.
4. Team skills and tooling tied to one platform.
5. Long-term contracts and commitments.

Q/ How to reduce lock-in ?

1. Containers and Kubernetes for portable workloads.
2. Infrastructure as Code (e.g. Terraform).
3. Open standards and portable data formats.
4. Regular exports and tested backups.
5. A written **exit plan** before you sign.

Security angle: lock-in is also an **availability and resilience** risk (outage or changed terms, and no quick alternative).

> **Hyperscale** : providers running data centers with very large server counts across a global footprint of **regions and availability zones**.

Q/ What makes a provider hyperscale ?

1. Data centers with very large server counts.
2. Global footprint of regions and zones.
3. Custom hardware, chips and networking.
4. Automation of almost everything.

> **Region** : a geographic area containing several availability zones.

> **Availability Zone** : a separate facility (one or more data centers) in a region, linked to the others by fast private networks, so **one failure doesn't take everything down**.

Security angle: **concentration risk** — much of the internet depends on a few hyperscalers, so one failure can ripple widely.

Inside the data center

> **Rack** : a standard frame (about **42U**) holding servers and a **top-of-rack switch**.

> **Aisle** : the space between rows, used to control cool and hot air flow (cold aisle in, hot aisle out).

> **Pod** : a group of racks sharing power, cooling and network uplinks — the unit you add as you grow.

Q/ What is the power path in a data center ?

- Utility grid → **UPS** (batteries bridge the gap) / backup **generator** → **PDU** (distributes power to each rack) → rack.

Q/ How is a data center cooled ?

1. Hot/cold aisle containment keeps air streams apart.
2. Air-conditioning units (CRAC/CRAH) move the heat out.
3. Chilled water and liquid cooling for very dense racks.
4. Site and climate choices can cut cooling cost.

> **PUE (Power Usage Effectiveness)** : total facility power ÷ power used by IT equipment. **1.0 is ideal**; lower = less energy wasted on cooling and power losses.

Example: facility uses 1.5 MW and IT equipment 1.0 MW → PUE = 1.5.

North-South traffic	East-West traffic
Between users outside and servers inside the data center	Between servers inside (replication, microservice calls, data processing)
Seen by the perimeter firewall	Usually the **larger** share; perimeter firewalls rarely see it

Security angle: attackers move **laterally** east-west; **micro-segmentation** and **zero trust** limit it.

> **Leaf-Spine** : every leaf switch connects to every spine switch, so any two servers are the **same number of hops** apart.

Q/ Why use leaf-spine ?

1. Predictable latency (equal hops).
2. More servers → add leaves; more bandwidth → add spines.
3. Many equal-cost paths: one failed link barely matters.

> **Fat-Tree** : the idea of many parallel equal paths with full bandwidth; leaf-spine is its common practical form.

> **Unified data center network** : one Ethernet fabric carries user, storage and management traffic instead of a separate network for each (fewer devices and cables, simpler operations; needs priorities and isolation).

Q/ What are the data-center security takeaways ?

1. Control physical access to racks and cages.
2. Separate the **management network** from user and storage traffic.
3. Watch east-west traffic, not only the perimeter.
4. Treat power and cooling redundancy as **availability controls**.

Security across deployment models

Model	Who secures what	Key risk	Typical controls
Private	The organization, all layers	Under-resourced patching and monitoring	Hardening, SIEM, physical access control
Public	Shared with the provider	Misconfigured storage and IAM	Least privilege, MFA, encryption, posture checks
Hybrid	Both sides, plus the link	Weak link, inconsistent policies	Encrypted tunnels, unified identity, segmentation
Multi-cloud	Each provider separately	Bigger attack surface, tool sprawl	Central identity, policy-as-code, unified logging

Rule of thumb: the more places your data lives, the more consistent your **identity and logging** must be.

Q/ Mini case — a national bank must keep records in the country, has unpredictable salary-day peaks, and fears one vendor. Which model ?

1. Regulated data → private or **hybrid**.
2. Unpredictable demand → public elasticity.
3. Fear of one vendor → **multi-cloud**.

- Recommendation: **hybrid** (records stay in a private/in-country cloud, the app front-end bursts to public), ideally with **more than one public provider**.

Summary of Cloud Computing L2: Types of Clouds, Providers & Data Centers. Built only from the lecture file; check the original slides for anything you're unsure about.